MPG No. 11.5.39 $4'/m'$ [Type III, tetragonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ T_{yzx} & -T_{xxz} & 0 \\ T_{xxz} & T_{yzx} & 0 \\ 0 & 0 & T_{yzx} \end{bmatrix}$$

$$T_{xxz} = \mathbb{T}_{z}^{\beta(1)}$$
$$T_{yzx} = \mathbb{T}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ T_{yzx} & \frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & T_{yzx} & 0 \\ 0 & 0 & -2T_{yzx} \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{M}_{xy}^{(1)}$$
$$T_{yzx} = \mathbb{M}_{v}^{(1)}$$
$$\frac{T_{xxz}}{2} = \mathbb{M}_{xy}^{(1)}$$
$$2T_{yzx} = 2\mathbb{M}_{v}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & T_{yzy} & 0 \\ T_{yzy} & -T_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzx} = \mathbb{M}_{v}^{(2)}$$
$$T_{yzy} = \mathbb{M}_{xy}^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).